



WMSU

INTRODUCTION TO HCI



Learning Objectives

At the end of the class, you can:

- Discuss the different definitions of HCI
- Apply the different principles, goals, importance of HCI, and its relationship to other disciplines
- Identify the different human factors in relation to HCI
- Exhibit awareness in the integration of HCI in the curriculum.

The top banner of the slide features a red background on the left with a white seal of the University of the Philippines, showing an eagle and the word 'UNIVERSITY'. On the right, there is a photograph of a large, yellow, multi-story building with arched windows, surrounded by green trees.

DEFINITIONS OF HCI



Human Computer Interaction is...

- Concerned with the design, evaluation, and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.

ACM Special Interest Group on
Human-Computer Interaction



Human Computer Interaction is...

- About designing computer systems that support people so that they can carry out their activities productively and safely.

Preece, et al.

As a Summary,

HCI	DESCRIPTION
HUMAN	A user which means an individual user, a group of users working together, or a sequence of users in an organization, each dealing with some part of the task or process.
COMPUTER	Any technology ranging from the general desktop computer to a large-scale computer system, a process control system or an embedded system.
INTERACTION	Any communication between a user and computer, be it direct or indirect.



HISTORY OF HCI

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- The term human–computer interaction has only been in widespread use since the early 1980s, but has its roots in more established disciplines
 - Systematic study of human performance began in earnest at the beginning of the last century in factories, with an emphasis on manual tasks.

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- The Second World War provided the impetus for studying the interaction between humans and machines, as each side strove to produce more effective weapons systems.
 - As computer use became more widespread, an increasing number of researchers specialized in studying the interaction between people and computers, concerning themselves with the physical, psychological and theoretical aspects of this process.

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- This research originally went under the name man-machine interaction, but this became human–computer interaction in recognition of the particular interest in computers and the composition of the user population



PRINCIPLES OF HCI



1. “Know Thy User”

- The term was coined by Hansen in 1971.
- The foremost creed in HCI is to devise interaction and interfaces around the **target users**.
- This principle simply states that the interaction and interface should cater to the needs and capabilities of the target user of the system in design.

2. Understand the Task

- the term task refers to the job to be accomplished by the user through the use of the interactive system.
- Understanding the task at hand is closely related to the interaction modelling and user analysis.

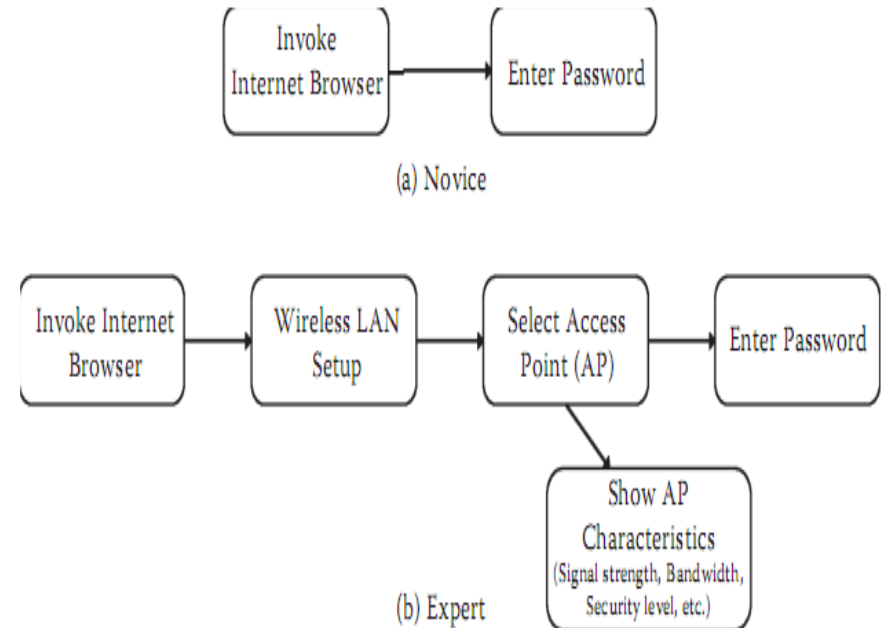


Figure 1.7 Two interaction models at different levels of detail for the task of “connecting to the Internet from a smartphone,” depending on the user type.

3. Reduce Memory Load

- Designing interaction with as little memory load
- Keeping the user's short-term memory load light is of particular importance with regard to the interface's role as a quick and easy guidance to the completion of the task



Figure 1.8 Interfaces designed for minimal short-term memory: (a) a menu system with fewer than 10 items (left) and (b) categorization by colors, areas, icons, and labels. Badges are used to display status information such as the current weather (see circled portions) and number of unread mails as a constant reminder. (From Microsoft®, Microsoft Metro interface, <http://www.microsoft.com>.)

4. Strive for Consistency

- In the longer term, one way to unburden the memory load is to keep **consistency**.
- consistency and familiarity also lead to higher acceptability and preference.
- For example, one way the Microsoft Windows®-based applications maintain their competitiveness is by promoting consistent and familiar interfaces



(a)

5. Remind Users and Refresh Their Memory

- The human memory dissipates information quite quickly, and this is especially true when switching tasks in multitasking situations (which is a very prevalent form of interaction these days).

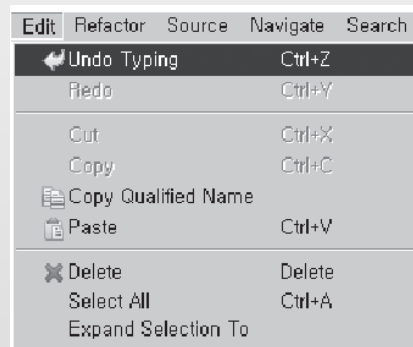


Figure 1.10 Reaffirming the user's action (i.e., credit card number correctly and securely entered) and a larger interactive process (i.e., the book purchase is complete).

6. Prevent Errors/Reversal of Action

- Error free operation is essential
- As such, the interaction and interface should be designed to avoid confusion and mental overload.

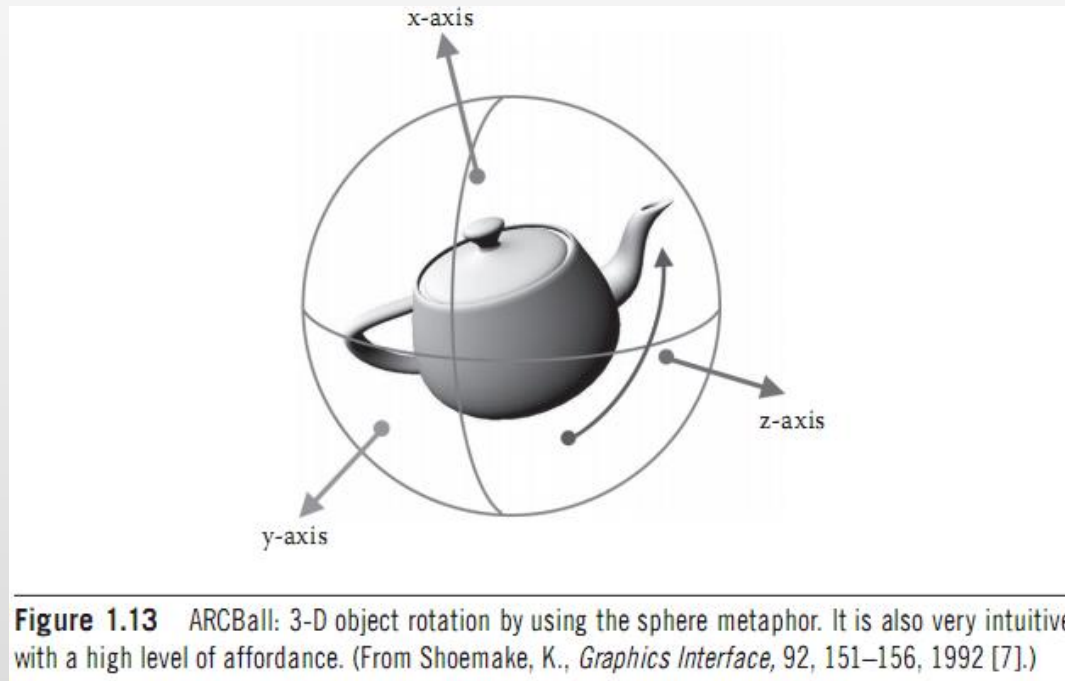
(a) Preventing errors by presenting only the relevant information at a given time (inactive menu items) and making selections rather than enforcing recall or full manual input specification.



(b) Making the user comfortable by always allowing an easy reversal of action.

7. Naturalness

- The final major HCI principle is to favor “natural” interaction and interfaces.
- Naturalness refers to a trait that is reflective of various operations in our everyday life.





GOALS OF HCI



1. Ensuring usability

“A usable software system is one that supports the effective and efficient completion of tasks in a given work context” (Karat and Dayton 1995).



The bottom-line benefits of more usable software system to business users include:

- Increased productivity
- Decreased user training time and cost
- Decreased user errors
- Increased accuracy of data input and data interpretation
- Decreased need for ongoing technical support



The bottom-line benefits of usability to development organizations include:

- Greater profits due to more competitive products/services
- Decreased overall development and maintenance costs
- Decreased customer support costs
- More follow-on business due to satisfied customers
- Not to use the term '**user-friendly**' which intended to mean a system with high usability but always misinterpreted to mean tidying up the screen displays to make it more pleasing



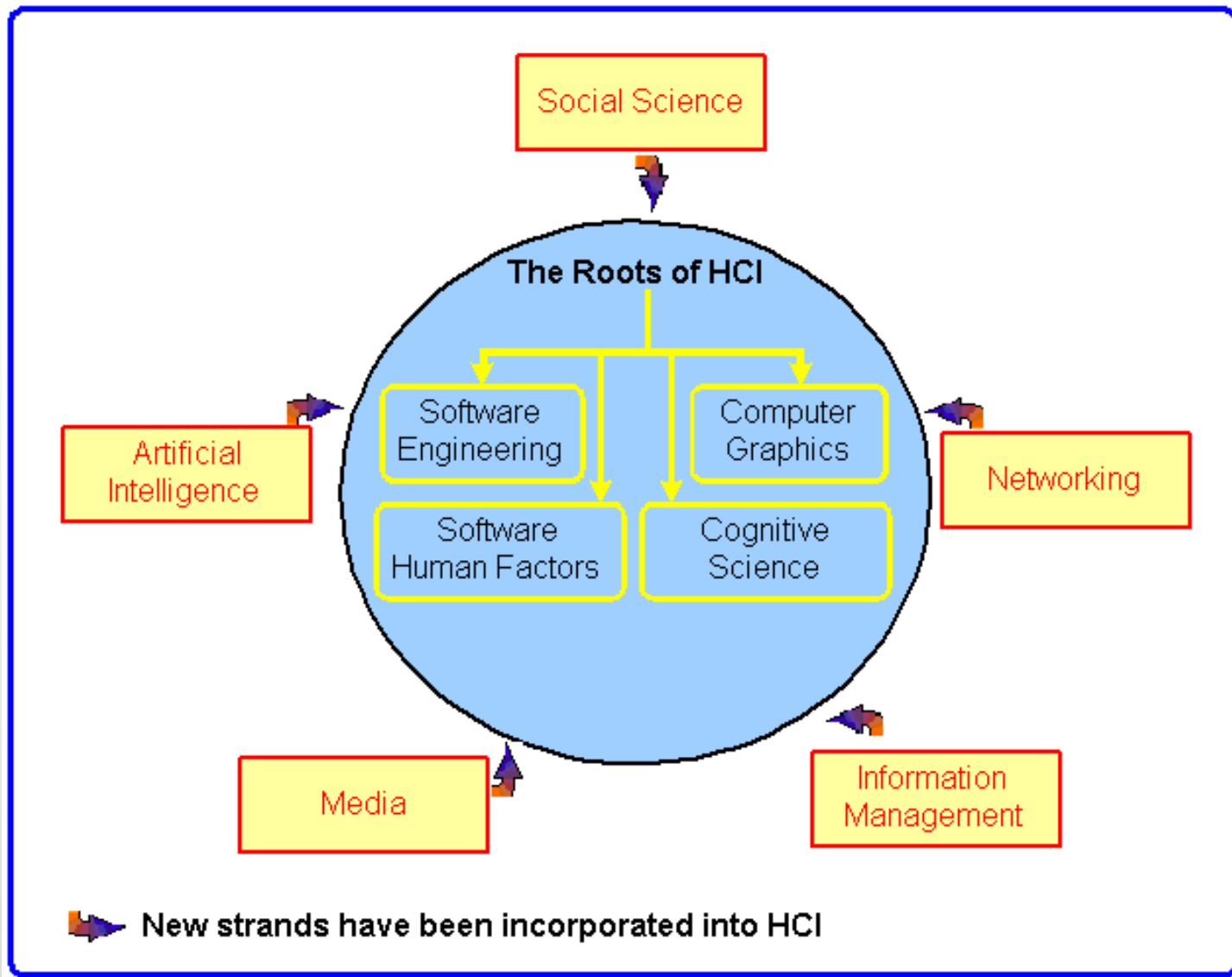
IMPORTANCE OF HCI

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- HCI will be increasingly important in the following areas:
 - As part of software development process and system design methods
 - As part of future legal requirements for software

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- As the basis for a set of usability criteria to evaluate and choose from amongst competing products
 - As the basis for successful marketing strategy to the increasingly important home and small business user



HCI RELATIONSHIP TO OTHER DISCIPLINES





Contributing disciplines

- Computer science
- Cognitive psychology
- Social and organizational psychology
- Ergonomics or human factors
- Linguistics
- Artificial intelligence
- Philosophy, sociology, anthropology
- Engineering and design



Cognitive Psychology

- Can help improve design by
 - providing knowledge about what users can and cannot expect to do
 - identifying and explaining the nature and causes of problems users encounter
 - supplying modeling tools and methods to help build interfaces that are easier to use



Social Knowledge

- Derived from social psychology, sociology, anthropology, linguistics, and philosophy
- Can help improve design by
 - providing knowledge about the context of use
 - identifying and explaining how people work together and what sorts of systems are needed to support collaborative work
 - supplying frameworks for social interaction and conversation that can form the basis of HCI frameworks



Organizational Knowledge

- Derived from organizational psychology, social psychology, sociology, management science
- Can help improve design by
 - providing models of the processes and structures of organizations
 - identifying trouble spots in organizations that are preventing computer systems from being used optimally and people from obtaining satisfaction from their work
 - supplying organizational models for the design and evaluation of new technologies



HUMAN FACTORS AND HCI

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- A science
 - derivation of principles of human behavior typically based on empirical testing using human subjects
 - A goal
 - To optimize human performance, including error reduction, increased throughput, user satisfaction and user comfort
 - Realization of computer field
 - *The user should not have to adapt to the interface*



HCI IN THE CURRICULUM

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- HCI involves both craft and science then it must, in part at least, be taught.
 - Presently, designers cannot afford to ignore the interface in favour of the functionality of their systems: the two are too closely intertwined.
 - HCI should be integrated into every computer science or software engineering course, either as a recurring feature of other modules or, preferably, as a module itself



GENDER AND HCI

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- Gender is an intricate part of everyday practice, affecting designers' interactions with one another, as well as the designs they create.
 - While recognizing differences is important to avoid homogenizing people, there has also been a tendency to rely on gender difference as a way to explain differences in numbers and usage in relation to technologies (Rode, 2011)

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- Gender awareness reflects on how gender may affect development practices and user interactions, explicitly recognizing the limitations of particular designs in terms of which genders or gendered characteristics are hindered and facilitated

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Activity

- Work on the workbook and answer the exercises 1 to 3 under Activity 1 – Introduction to HCI.



REFERENCES

- Dix, A. Finlay J., Abowd, G, and Beale, R. (2004) Human Computer Interaction 3rd ed., Pearson Education Limited
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